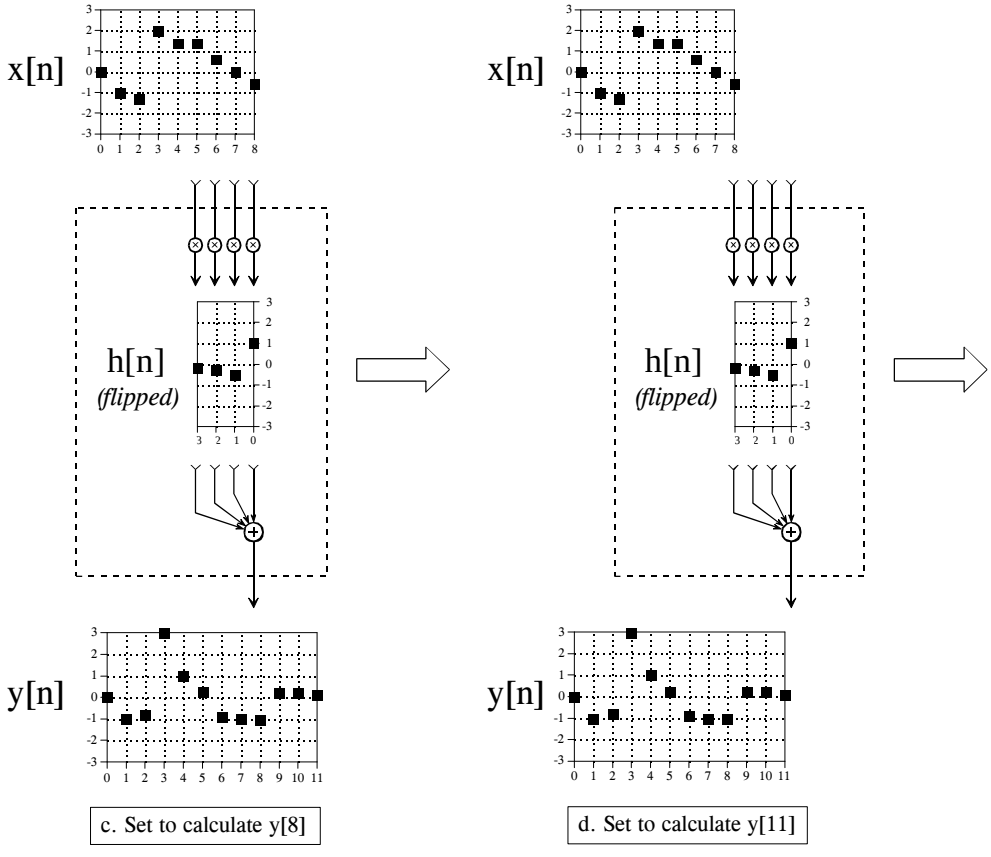

Figure 6-9 (continued)

The important part is that the far left and far right samples in the output signal are based on *incomplete* information. In DSP jargon, **the impulse response is not fully immersed in the input signal**. If the impulse response is M points in length, the first *and* last $M-1$ samples in the output signal are based on less information than the samples between. This is analogous to an electronic circuit requiring a certain amount of time to stabilize after the power is applied. The difference is that this transient is easy to ignore in electronics, but very prominent in DSP.

Figure 6-10 shows an example of the trouble these end effects can cause. The input signal is a sine wave plus a DC component. The desire is to remove the DC part of the signal, while leaving the sine wave intact. This calls for a high-pass filter, such as the impulse response shown in the figure. The problem is, the first and last 30 points are a mess! The shape of these end regions can be understood by imagining the input signal padded with 30 zeros on the left side, samples $x[-1]$ through $x[-30]$, and 30 zeros on the right, samples $x[81]$ through $x[110]$. The output signal can then be viewed as a filtered version of this longer waveform. These "end effect" problems are widespread in